

CHAT MODEL API CHEATSHEET

AI Engineer Ready · Week 3 — invoke / batch / stream

Golden Rule → one prompt = invoke · many independent prompts = batch · progressive output = stream

1. Setup (Once Per Session)

```
from langchain.chat_models import init_chat_model
from langchain.messages import SystemMessage, HumanMessage

chat = init_chat_model(model="gpt-4o-mini", model_provider="openai")
```

Reads `OPENAI_API_KEY` automatically from your environment (via `load_dotenv()`). `model_provider` lets the same code work with other providers later.

2. invoke — One Prompt In, One Response Out

Use when: a single request, and you need the answer before moving on.

```
resp = chat.invoke([
    SystemMessage(content="You are a concise assistant."),
    HumanMessage(content="Summarize LangChain in one paragraph."),
])
print(resp.content)

# A plain string also works — treated as a single user message
resp = chat.invoke("Explain LangChain in one sentence.")
```

3. batch — Many Prompts In, Many Responses Out

Use when: several independent prompts, processed together (more efficient than looping invoke).

```
prompts = ["Define LangChain.", "Define a prompt template.", "Define a chat model."]
responses = chat.batch(prompts)
for p, r in zip(prompts, responses):
    print(p, "->", r.content)
```

Order of `responses` matches order of `prompts`.

4. stream — Response Arrives in Chunks

Use when: showing output progressively (typing effect) or watching a long generation happen.

```
for chunk in chat.stream("Write a short poem about learning to code."):
    print(chunk.content, end="", flush=True)
```

5. Quick Comparison

	INVOKE	BATCH	STREAM
Input	1 prompt	list of prompts	1 prompt

	INVOKE	BATCH	STREAM
Output	1 response, returned when ready	list of responses, all together	response arrives in pieces
Use for	Simple single call	Many independent calls	Progressive / UX display
Get text with	resp.content	[r.content for r in responses]	chunk.content per chunk

Which One Should Beginners Start With?

invoke. Simplest mental model — one call, one answer — and matches how you debug most issues. Move to batch once you have multiple independent prompts, and stream once latency/UX matters.

6. Extracting Content Safely

```
resp = chat.invoke(messages)
text = getattr(resp, "content", str(resp)) # won't crash if shape is unexpected
print(text)
```